Coproduct.

Def. Let $\{X_j \mid j \in J\}$ be a collection of Topological Spaces.
Consider the set $\coprod_{j\in J} X_j \overset{\text{def}}{=} \bigcup_{j\in J} X_j \times \{j\}$
and for each $i \in J$, Consider the injective map

$$\varphi_i : X_i \to \coprod_{j\in J} X_j : x \mapsto (x, i).$$

The Topology $\mathcal{U}$ on $\coprod_{j\in J}$ which is the largest Topology such that for all $j \in J$, φ_j is Continuous is called the disjoint union topology.
Simply, denote $\coprod X_j$.

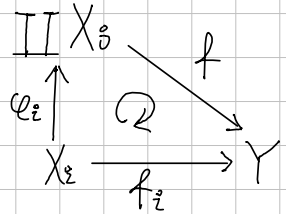
Prop. Let $U \subset \coprod X_j$ be a subset of the disjoint union space. Then,
U is open set of $\coprod X_j$ if and only if for all $i \in J$, $\varphi_i^{-1}[U] \subseteq X_i$ is open.

Proof. Since the Topology on $\coprod X_j$ is the final Topology by $\{\varphi_i \mid i \in J\}$, clear. ∎

Diagramically, for given a space Y and Continuous maps $f_j : X_j \to Y$ $(j \in J)$,
there exists a unique Continuous map

$$f : \coprod X_j \to Y : (x, j) \mapsto f_j(x)$$

which satisfies that for all $j \in J$, $f_j = f \circ \varphi_j$.



Adjunction space.

Def. Let X and Y be Topological spaces, and $A \subset X$ be subspace. For a Continuous map $f: A \to Y$, define a equivalent relation on $X \amalg Y$ which is generated by for all $a \in A$, $a \sim f(a)$.

Then, the quotient topology

$$X \cup_f Y \overset{\text{def}}{=} (X \amalg Y)/\sim$$ is called the adjunction space by f.

And, the map f is called the attaching map.

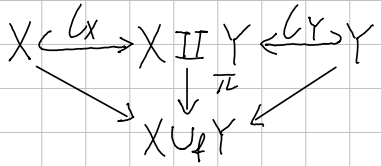
Note: Specifically, the equivalent relation $\sim$ on $X \amalg Y$ is consisted by:

$$[z] = \begin{cases} f^{-1}[f[[z]]] \cup f[[z]] & \text{if } z \in A \\ \{z\} & \text{if } z \in X \setminus A \\ f^{-1}[[z]] \cup \{z\} & \text{if } z \in f[A] \\ \{z\} & \text{if } z \in Y \setminus f[A] \end{cases}$$

In language of function, define the canonical embbedings

$\iota_X : X \to X \sqcup Y$ and $\iota_Y : Y \to X \sqcup Y$

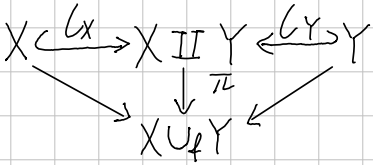
$: x \mapsto (x, 0)$ $: y \mapsto (y, 1)$



Lemma. Let $A \subset X$ be a closed (or open) set, and $f: A \to Y$ be continuous. Then, the quotient map $\pi: X \amalg Y \to X \cup_f Y$ induces a Homeomorphism

$$\pi|_Y : Y \xrightarrow{\cong} \pi[Y] \quad \text{(Canonical Copy)}$$

Moreover, $\pi[Y] \subseteq X \cup_f Y$ is a Closed (or open) set.



Examples.

Mapping Cylinder.

Let $f: X \to Y$ be a Continuous map. Then, for an interval $I = [0, 1]$, let $\tilde{f}: X \times \{1\} \to Y : (x, 1) \mapsto f(x)$. Define the adjunction space

$$M_f \overset{\text{def}}{=} (X \times I) \cup_{\tilde{f}} Y$$